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28 March 1962
SUPERSEDING
MIL-D-16211C
16 November 1954

MILITARY SPECIFICATION

DUMMY LOAD, ELECTRICAL, DUMMY ANTENNA TS-235B/UP

This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers Electrical Dummy Load, Dummy Antenna TS-235B/UP. (See 6.1.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-C-71 - Connectors, "N", for Radio Frequency Cables.
MIL-P-116 - Preservation, Methods of.
MIL-C-3432 - Cable and Wire, Electrical Power and Control, Flexible and Extra Flexible, 300 and 600 Volts.
MIL-C-5015 - Connectors, Electrical, "AN" Type
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts; Preparation for Delivery of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. Dummy loads furnished under this specification shall be a product which has been tested and has passed the preproduction inspection specified in 4.4.

3.2 Material. Material used in the construction of the equipment shall be such as to enable it to meet the requirements of this specification.

3.3 Design and construction. The equipment shall consist of a coaxial structure, 1-5/8 inches in diameter, containing a dissipative dielectric and provided with radiating fins, cooling housing, and input connector incorporating a probe.

3.3.1 Fan. When necessary to meet the requirements specified herein a fan with cable for connection to an external power supply shall be provided. The fan shall be within the cooling housing and shall be mounted in such a manner as to permit the entire unit to meet all the requirements of this specification.

3.3.1.1 Fan motor. The fan motor shall operate from an external 115-volt, 400-cycle power supply. A cable, type MG in accordance with Specification MIL-C-3432, for connecting the fan motor to the power supply shall be permanently fastened to the unit at one end, and shall be equipped with an AN-3106A-14S-2P connector in accordance with Specification MIL-C-5015, at the other end.

3.3.2 Input connector. The input connector shall provide direct coupling of the load to a 50 ohm, 1-5/8-inch coaxial line. The mating dimensions for the coupling shall be such that it mates with the dimensions shown on figure 1. A protective cap shall be attached to the connector for appropriate use. A removable adapter for the coaxial line center conductor shall be incorporated in the input connector. The contact of the center conductor adapter shall meet the gage requirements of figure 2. When completely engaged in the load, the center conductor shall be centrally located within the outer conductor. A receptacle with protective cap shall be provided for storage of the adapter in the equipment cooling housing.

3.3.2.1 Probe coupling loss. The input connector shall incorporate a probe to enable coupling of other test equipment into the coaxial transmission line. The loss due to this probe shall be 50±0 5

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decibles (db) at 1300 mc., and shall not vary more than +5 -2 db from this value over the frequency range of 500 to 1500 mc and operating temperature range when tested as specified in 4.6.2. The probe termination shall be series N jack in accordance with Specification MIL-C-71.

3.3.3 Transit and combination case. Neither a transit case nor a combination type case will be required. Two rigid, metal handle guards shall be designed into the dummy load panel in such a way that the receptacle, with protective cap attached, shall not extend beyond the handles.

3.3.4 Dimensions and weight. Dimensions of the equipment shall not exceed 24 by 4-7/8 by 6-3/4 inches. The total weight shall not exceed 20 pounds.

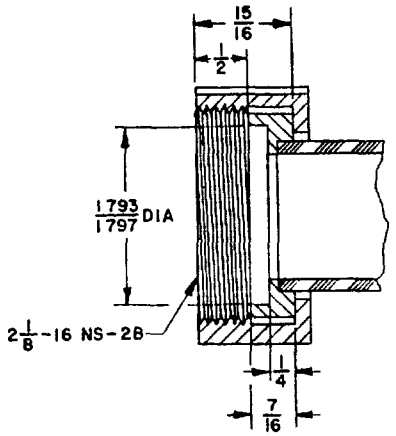


Figure 1. Coupling mating dimensions.

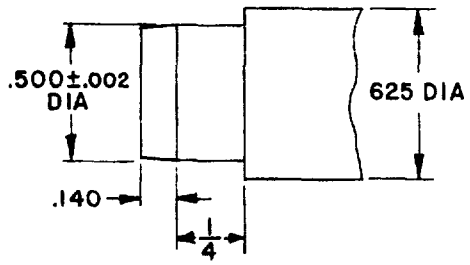


Figure 2. Center conductor gage.

3.4 Voltage standing wave ratio. The voltage standing wave ratio (VSWR) of the load when used as termination for a 50-ohm, 1-5/8-inch coaxial line shall not exceed 1.30 to 1 to 500 mc. and 1.2 to 1 over the frequency range of 600 mc. to 1500 mc. (See 4.6.3.)

3.5 Power dissipation. The equipment shall be capable of dissipating pulse radio frequency (r.f.) power having an average value of 1,000 watts and a duty cycle of 0.0008. The temperature on the surface of the cooling case shall not exceed 76° C. after the load has been operating for 1 hour at maximum rated power. (See 4.6.4.) The temperature of the coupling probe and coupling nut may exceed the temperature specified herein.

3.6 Radio interference reduction. The load under full rated load at any frequency within its specified operating frequency range shall not exhibit any indication of radiated interference in any direction when measured as specified in 4.6.5.

3.7 Shock and vibration. There shall be no visible evidence of damage and the load shall meet the requirement for VSWR (see 3.4) and probe coupling loss (see 3.3.2.1) after being subjected to shock and vibration. (See 4.6.6.)

3.8 Instruction pamplets. Two instruction pamphlets shall be furnished with each dummy antenna.

3.9 Marking. The equipment shall be marked in accordance with Standard MIL-STD-130. The following information shall be placed on the identification plate (see 6.2):

(a) Nomenclature.
(b) Manufacturer's identification.
(c) Contract or order number.
(d) Special characteristics.

3.10 Workmanship. Loads shall be manufactured and processed in such a manner as to be uniform in quality. The exterior of the loads shall be free from tool marks, burrs and deep scratches.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of

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sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 Classification of inspection. The examination and testing of loads shall be classified as follows:

(a) Preproduction inspection. (See 4.4.)
(b) Acceptance inspection. (See 4.5.)
1. Inspection of Product for delivery. (See 4.5.1.)
2. Inspection of preparation for delivery. (See 4.5.2.)

4.3 Inspection conditions. Unless otherwise specified herein, all inspection shall be made at room ambient temperature, pressure and humidity.

4.4 Preproduction inspection. Preproduction inspection shall be performed by the supplier, after award of contract and prior to production, at a location designated by the Government. (See 6.2.)

4.4.1 Sample. The sample submitted for preproduction tests shall be representative of the manufacturer's normal production. Two units of product shall be provided for preproduction inspection.

4.4.2 Inspection routine. The specimens shall be subjected to the examinations and tests specified in table I, in the order shown.

4.4.3 Failure. Failure in any examination or test will be cause for rejection of the preproduction models.

Table I - Preproduction inspection.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical examination	3.2 to 3.3.2 incl, 3.3.3, 3.3.4, 3.8, 3.9, 3.10	4.6.1
Probe coupling loss	3.3.2.1	4.6.2
VSWR	3.4	4.6.3
Power dissipation	3.5	4.6.4
Radio interference	3.6	4.6.5
Shock and vibration	3.7	4.6.6

4.4.4 Reports. Five copies of test reports shall be submitted to the contracting officer of the agency responsible for the procurement. The report shall describe the test procedure, test equipment and results. The results of tests shall be certified by a responsible official of the manufacturing concern.

4.5 Acceptance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A, B, and C.

4.5.1.1 Inspection lot. An inspection lot, as far as practicable, shall consist of all loads produced under essentially the same conditions and offered for inspection at one time.

4.5.1.2 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot, rework it to correct the defects, or screen out the defective units, and again inspect it, as applicable. Such lots shall be kept separate from new lots and shall be clearly identified as reinspected lots. Such lots shall be reinspected, using tightened inspection.

4.5.1.3 Group A inspection. Group A inspection shall consist of visual and mechanical examination (see 3.2, 3.3, 3.3.1, 3.3.1.1, 3.3.2, 3.3.3, 3.3.4, 3.8, 3.9 and 3.10) and shall be made on the same set of sample units.

4.5.1.3.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for ordinary inspection. The acceptable quality level (AQL) shall be 1.0 and 4.0 (percent defective) for major and minor defects, respectively. Major and minor defects shall be as defined in Standard MIL-STD-105.

4.5.1.4 Group B inspection. Group B inspection shall consist of the tests specified in table II.

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Table II - Group B inspection.

Examination or test	Requirement paragraph	Method paragraph
Probe coupling loss	3.3.2.1	4.6.2
VSWR	3.4	4.6.3
Power dissipation	3.5	4.6.4

4.5.1.4.1 Sampling plan. The sampling plan shall be in accordance with Standard MIL-STD-105 for small sample inspection. Unless otherwise specified herein normal inspection shall be used at the start of the contract. For small sample reduced inspection procedure R-1 shall be used. The inspection level shall be L-8 for normal and tightened inspection and L-6 for reduced inspection. The AQL shall be 4.0 (percent defective) for all tests combined.

4.5.1.5 Group C inspection. Group C inspection shall consist of the tests specified in table III. Shipment shall not be held up pending results of the inspection.

4.5.1.5.1 Sampling plan. Three sample units shall be selected from each 300 dummy loads or fraction thereof produced during each three-month interval, except that three sample units shall be selected from the first 30 dummy loads produced.

4.5 1 5.2 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or order.

4.5.1.5.3 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or process, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc, and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspection may be reinstituted, however, final acceptance shall be withheld until the group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information, concerning the failure and the corrective action taken shall be furnished to the contracting officer

4.5.2 Inspection of preparation for delivery. Sample items and packs shall be selected and inspected in accordance with Specification MIL-P-116 to verify conformance with requirements in section 5 of this specification.

4.6 Methods of examination and test.

4.6.1 Visual and mechanical examination. Dummy loads shall be examined to verify that materials, (see 3.2) design and construction, (see 3.3 to 3.3.4 incl and 3.8), mating dimensions, (see 3.3.2) marking, (see 3.9) and workmanship (see 3.10) are in accordance with the applicable requirements.

4.6.2 Probe coupling loss. The probe coupling loss shall be measured by any suitable method at 3 frequencies within the band of 500 to 1500 mc. in addition to the 1300 mc. reference frequency. (See 3.3.2.1.)

4.6.3 VSWR. The VSWR of the loads shall be measured by a suitable method in 100 mc. steps from 500 to 1500 mc. (See 3.4.)

4.6.4 Power dissipation. The loads shall be subjected to the average power specified with required duty cycle for 10 cycles of 1 hour power on and a minimum of 1 hour with power off at any frequency within the specified frequency range. The temperature at the surface of the transmission line coupling and cooling housing shall be measured after the load reaches thermal equilibrium by means of a thermal coupler. At the end of the tenth cycle, the VSWR of the load shall be measured. (See 3.5.)

4.6.5 Radio interference reduction. Radio interference reduction shall be determined by using a Radio Test Set AN/URM-17 with its dipole antenna tuned to resonance at the operating frequency of the dummy load between 500 and 1000 mc. Any available receiver shall be used from 1000 to 3000 mc. The antenna for all tests shall be located 3 feet from the dummy load (See 3.6.)

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Table III. —Group C inspection.

Test	Requirement paragraph	Method paragraph
Radio interference reduction	3.6	4.6.5
Shock and vibration	3.7	4.6.6

4.6.6 Shock and vibration. (See 3.7.)

4.6.6.1 Shock. Loads shall be tested in accordance with Standard MIL-STD-202, using method 205B, test condition B. The following details and exceptions shall apply:

(a) Tests and measurements prior to shock - Not applicable
(b) Method of mounting - Position normal to use
(c) Number of blows - Three
(d) Measurements after shock - VSWR (see 3.4) and Probe coupling loss (see 3.3.2.1)

4.6.6.2 Vibration. Loads shall be tested in accordance with Standard MIL-STD-202, using method 201A. The following details and exceptions shall apply:

(a) Tests and measurements prior to vibration - Not applicable
(b) Method of mounting - Position normal to use
(c) Measurements after vibration - VSWR (see 3.4) and Probe coupling loss (see 3.3.2.1)

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. The equipment and instruction pamphlets shall be prepared for domestic shipment, immediate use; domestic shipment and storage, or overseas shipment, as specified (see 6.2), in accordance with Specification MIL-E-17555. Method II packaging shall be used for the domestic shipment and storage and overseas shipment levels.

6. NOTES

6.1 Intended use.

6.1.1 Function. The equipment is meant to perform the following functions:

a. Act as a nonreflecting load for terminating 50-ohm 1-5/8-inch coaxial line.
b. Provide a terminated probe coupling through which other test equipment can be connected to the transmission line.

6.1.2 Frequency range. The TS-235/UP is intended to operate over the frequency range of 500 to 1500 megacycles.

6.2 Ordering data. Procurement documents should specify the following.

(a) Title, number, and date of this specification.
(b) Where preproduction tests are to be conducted (see 4.4).
(c) Whether delivery shall be for domestic shipment, immediate use; domestic shipment and storage, or overseas shipment (see 5.1).
(d) Identification plate data shall reference AN/TPS-1( ) Radar as the equipment with which the dummy antenna is used. (See 3.9.)

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

INSTRUCTIONS

BUDGET BU. NO. 45-R30

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING

b RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES" IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification.)* PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU

SUBMITTED BY *(Print name and activity)*	DATE

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